

TOPICAL REVIEW

A Review of Lightweight IoT Authentication Protocols From the Perspective of Security Requirements, Computation, Communication, and Hardware Costs

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ABSTRACT The Internet of Things (IoT) is a broad area that encompasses various use cases. Numerous use cases include various IoT devices, which necessitate various cryptographic methods to provide distinct security requirements. Due to their limited resources, most IoT devices require lightweight cryptographic implementations. The requirements of use cases must be revealed to use cryptography effectively. Therefore, classifying the security requirements for these scenarios is beneficial when blueprinting a new system design. Authentication is the first level of security and is crucial for ensuring the secure exchange of information. Designing an IoT authentication protocol requires addressing both security requirements and physical constraints. This paper reviews and classifies the studies in IoT authentication published in the last decade. The studies are categorized into two main categories: the authentication factors and commonly preferred cryptographic primitives. Authentication factors include passwords, RFID, smart cards, and OTPs, while cryptographic primitives focus on techniques such as ECC and PUFs. Following the study selection, this review evaluates the provided security requirements alongside computational, communication, and hardware costs, while also considering application domains. It concludes by highlighting the most common approaches and identifying existing gaps in the literature. Additionally, it examines the trade-offs between attack resistance and protocol costs.

INDEX TERMS Authentication survey, authentication review, Internet of Things, IoT security, lightweight authentication, IoT authentication.

I. INTRODUCTION

The Internet of Things (IoT) is the concept of a computer system that employs heterogeneous collaboratively working computers with sensing, actuating, and networking capabilities to provide a service to its users. IoT adoption increases and presents exciting opportunities but also necessitates carefully considering security measures. Furthermore, most IoT deployments opt for low-cost devices that are constrained resourcewise. This adds another challenge to satisfy security requirements properly. The rapid increase in low-cost IoT

deployment underscores the critical importance of robust lightweight authentication protocols. When designing an IoT application, it's crucial to establish its purpose and the conditions under which it will operate. At the same time, the IoT application security mechanism should be determined according to the requirements of the desired usage scenario. Nevertheless, every scenario that requires device management necessitates a dependable authentication mechanism as a foundation for the upcoming security measures. Authentication is a protocol that verifies whether the interacting party is a pre-registered computer system user [1]. Depending on the use case, an authentication protocol might require physical or remote interaction of the

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user. Therefore, numerous use cases lead to numerous authentication protocols with distinct motivations and goals. These studies could be classified based on security requirements and costs. The provided security requirements and features of the studies could be discussed based on cryptographic primitives, application areas, spatiality of access, and authentication factors. Costs could be reviewed based on computation, communication, and hardware usage. Thus, whether the studies could be sufficient and also efficient for IoT devices.

This study reviews the literature on IoT authentication protocols based on multiple perspectives. First, authentication factors and cryptographic primitives that are frequently used are selected. After that, the study examines the cryptographic primitives that are commonly preferred, highlighting the most utilized techniques in IoT authentication. Additionally, it assesses the costs associated with these protocols regarding computation, communication, and hardware usage providing a comprehensive analysis of their efficiency. This establishes a relationship between the security of protocols and their associated costs. Finally, the study evaluates the strengths of the protocols and security solutions and shows their effectiveness in mitigating potential security breaches. These findings highlight the cost requirements needed to meet specific security requirements. Through these questions, the study aims to provide a detailed and structured review of the current state of IoT authentication protocols. The results are presented in tables to guide the design of IoT protocols tailored for resource-constrained devices.

The contributions of the study are the following.

- The selected studies are classified based on the classification that is shown in Section II.
- The studies are reviewed based on the cryptographic primitives in Section III-A.
- The studies are reviewed based on the authentication factors in Section III-B.
- Section III-C reviews the studies based on specific application areas.
- Section III-D examines the strengths of the studies against various attacks as claimed by the studies.

The tables compare various aspects of the studies, including their characteristics, overheads, other methods and authentication factors that are used, security measures against attacks, computation, communication, and hardware costs. Section IV presents the review and survey studies that classify IoT authentication studies. The final section, Section V, summarizes the conclusions of this study. Table 17 presents an overall comparison of the selected studies based on their security requirements and costs.

A. RESEARCH QUESTIONS

This review is structured around the following research questions.

- What are the mostly preferred IoT authentication factors?
- Which cryptographic primitives are utilized in IoT authentication?

- What is the communication cost of the protocols?
- What is the computation cost of the protocols?
- Is there a correlation between attack resistance and the cost of the protocols?
- What are the most common application areas for these protocols?

B. REVIEW METHODOLOGY

This review leverages prominent electronic databases to select papers on IoT authentication. The databases include ACM Digital Library, IEEE Xplore, ScienceDirect, Springer, and Wiley Online Library. Figure 1 illustrates the utilized libraries, search criteria, and features of the selected papers.

The inclusion criteria require that the papers provide cost-related information (at least one aspect) and focus on studies involving IoT devices.

II. EVALUATION CRITERIA AND CHARACTERISTICS

The studies could be classified as in Figure 2 [2], [3], [4], [5]. This classification has two perspectives and the paper selection of this study is based on this classification. While the sub-categories can be further expanded, the categories highlighted in red represent the main focus of this paper.

The selected studies are reviewed into two main axes which are primitives (lightweight cryptographic methods) and authentication factors as shown in Table 1. The table shows the intersection of these categories, and it is seen that most of the studies are only in one category.

- **Security Requirements:** This study focuses on the security requirements and features of the selected studies. The selected studies are reviewed based on their authentication factors and the cryptographic methods that are preferred to provide the requirements.
 - **Authentication Factor:** Authentication relies on three main factors: knowledge, possession, and inherence. *Knowledge factor* refers to something the user knows; such as passwords, pin codes, and answers to some questions. *Possession factor* involves something the user has; such as smart cards and RFID tags. *Inherence factor* refers to something the user is; such as fingerprints and retina patterns. This study classifies the selected literature studies based on the mentioned types of factors. Some of the authentication methods of the factors are predominantly used in the literature. In this survey, smart card and RFID methods from possession factors; password method from knowledge factors; OTP method from knowledge, and possession factors (two-factor authentication) are reviewed. Figure 2 shows the authentication factor classification and the selected methods for this study. Sections include literature studies that are classified **based on their main authentication factor**. While some studies focus on a single factor, most of them utilize two-factor authentication.

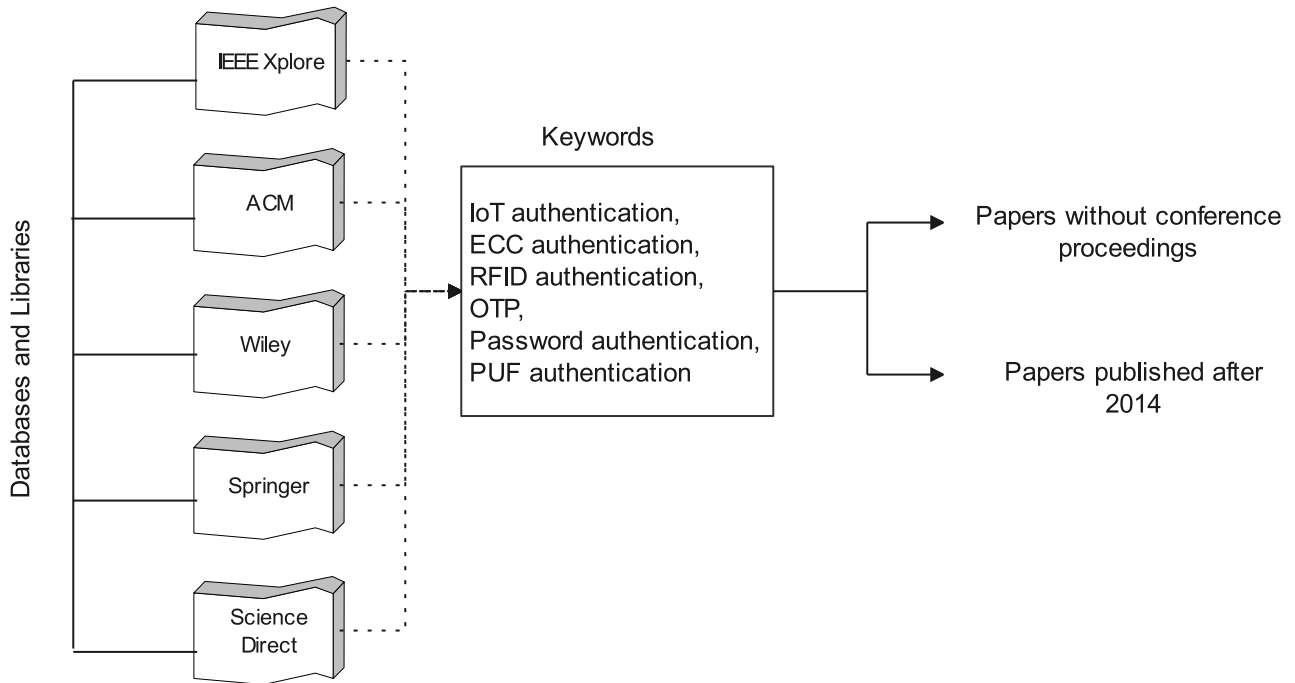


FIGURE 1. The papers were collected from digital databases and libraries, focusing on those published after 2014, with an emphasis on more recent studies.

- Cryptographic Primitives: Authentication protocols include various cryptographic primitives, functions, and operations. IoT devices primarily use lightweight methods due to their limited resources. Authentication protocols, including ECC and PUF primitives, are reviewed due to their frequent usage.

Several studies have been selected to be reviewed in this paper and are evaluated based on the above criteria, with preference given to recent studies. The selected studies are also evaluated based on their security requirements, cost, usage areas, and strengths.

- Cost: Most IoT devices are resource-constrained, IoT authentication mechanisms should be low-cost. Cost is one of the critical parameters for IoT authentication protocols. This review categorizes the cost parameter as follows to evaluate the studies:
 - Communication Cost: The selected studies are evaluated based on their computation loads, with the authors specifying the costs of their studies in terms of bytes or bits.
 - Computation Cost: The studies are also summarized based on their functions and operations. Each study involves specific computations, including several cryptographic functions and operations. In this review, each classification includes relevant functions, evaluated separately in their respective sections.
 - Additional Hardware Cost: Additional hardware such as smart cards and their readers, hardware that enables PUF, and RFID tags usage increase the overall cost of the authentication mechanism.

- Usage: In this review, the studies are evaluated based on their application areas. There could be various application areas of IoT devices such as autonomous aerial vehicles, health systems, wireless body area networks (WBAN), and wireless sensor networks (WSN).
- Security/Robustness: This paper considers the claimed mitigation of common attacks. The reviewed studies are evaluated based on their security requirements against enlisted attacks. Each study aims to mitigate a list of attacks. For an objective review, the strength of the studies is shown relative to the attacks for which the related studies specify that they provide protection against the reviewed attack types.

III. AUTHENTICATION MECHANISMS FOR IoT DEVICES

This section evaluates studies of authentication mechanisms from two perspectives. The first perspective is the cryptographic primitive used in the studies. The second perspective is the commonly used authentication factors.

A. CRYPTOGRAPHIC METHODS USED IN AUTHENTIFICATIONS

Many cryptographic primitives are used to improve the security of devices. The authentication protocols include cryptographic hash functions and XOR operations. In addition, two primitives are used frequently. The first primitive is Elliptic Curve Cryptography (ECC) and the second is Physically Unclonable Functions (PUFs). The primitives are analyzed with the literature studies in terms of their

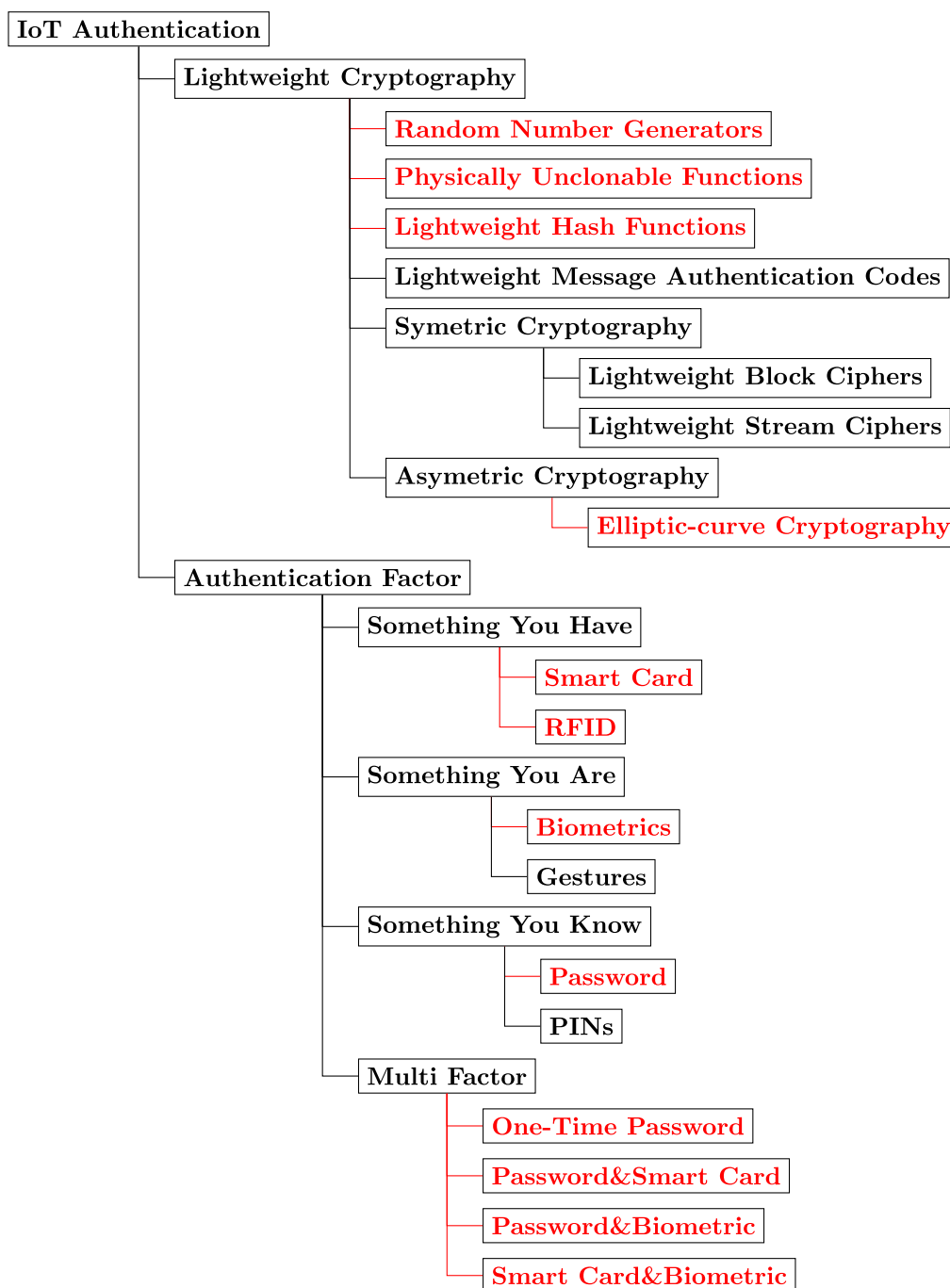


FIGURE 2. IoT authentication protocols are classified into two main categories: lightweight cryptographic primitives and authentication factors. Lightweight cryptography is further categorized based on the most commonly used primitives, while authentication factors are grouped according to the most frequently preferred methods. This study specifically focuses on the categories highlighted in red.

characteristics including costs and authentication factors used.

1) ELLIPTIC CURVE CRYPTOGRAPHY

Elliptic Curve Cryptography (ECC) is one of the most commonly used primitives in lightweight authentication

protocols, owing to its efficient computation and small key sizes. ECC is a lightweight public-key cryptographic function, making it highly suitable for resource-constrained devices. It offers a level of security comparable to RSA or discrete logarithm systems but achieves this with significantly fewer computations, making it both efficient and practical for environments with limited processing

TABLE 1. The reviewed studies are selected based on the use of cryptographic primitives, authentication factors, or a combination of both.

		Utilized Cryptographic Primitive		
		ECC	PUF	Other
Authentication Factors	Password	[6], [7], [8], [9], [10]	[11]	[12], [13], [14], [15], [16], [17], [18], [9], [19]
	RFID	[20], [21], [22]	[23]	[24], [25], [26], [27], [28], [29], [30], [31], [32], [33]
	Smart Card	[34]	–	[35], [36], [37], [38], [39], [40], [41], [42], [43], [44]
	OTP	[45], [46]	–	[47], [48], [49], [50], [51], [52]
	Other Multi Factor	[53], [54]	[55]	–
	Other	[56], [57], [58]	[59], [60], [61], [62], [63], [64], [65], [66]	[67], [68], [69], [70], [71], [72], [73], [74], [75]

power and energy resources [3]. Table 2 summarizes the studies concerning their advantages, authentication types, and communication costs. Table 3 compares the studies concerning their computation cost.

Liao and Hsiao [20] propose an ECC-based RFID authentication protocol, including designing an ID-verifier transfer protocol. The protocol uses a challenge-response protocol to secure the exchange of messages. The authors provide a secure transmission with an ID-verifier protocol between the tag and the reader.

Another study [6] proposes an ECC-based mutual authentication that uses HTTP cookies. The protocol provides communication security between embedded devices and cloud servers.

Rostampour et al. [56] propose an ECC-based lightweight authentication for IoT edge devices. The authors only use ECC function to decrease the complexity. Traceability and Man in the Middle attacks are focused.

Li et al. [45] propose an ECC-based authentication and key exchange protocol. The authentication information (identity and password) is updated each session using the ECC method. The authors present a more performing algorithm than similar studies.

Iqbal et al. [57] propose an ECC-based authentication protocol between IoT devices and cloud servers. The authors prefer the extended Canetti–Krawczyk (eCK) adversary model instead of the Dolev–Yao model because they claim that the eCK model is more strict for authentication protocols.

Singh et al. [53] propose an ECC-based mutual authentication and key agreement protocol that includes timestamps to identify sessions. The authors use passwords and biometrics for the login phase between the user and the mobile device while the authentication and key establishment phases are between the user, the node, and the gateway, separately.

He et al. [7] use bitwise XOR, symmetric encryption, and ECC to strengthen the authentication and key exchange. Timestamps are used to identify sessions.

As a different approach to ECC-based authentication, Patel and Doshi [8] propose a level-dependent authentication protocol to reduce the authentication complexity. In this

approach, users can authenticate with their lower-level devices, which also minimizes the number of registration and initialization phases. The protocol includes only XOR operations, hash functions, and ECC encryption with low computation cost. The protocol provides multiple registrations of a user for various IoT devices.

Jayashree and Santhosh Kumar [58] propose a lightweight ECC-based authentication protocol for VANETs. The authors provide a multi-level authentication between devices, road-side infrastructure, and the authentication server. The communication messages are signed and the communication delay is decreased by improving the signature algorithm.

Abbasi et al. [54] propose a lightweight ECC-based secure authentication protocol for healthcare systems. The biometric authentication factor enhances the protocol's security.

As a distinct approach for IoT authentication, Saqib and Moon [10] propose a protocol that aims to provide security for MQTT-based IoT applications. The protocol includes a low-cost signature, fuzzy extractor, biometric, and ECC. The session key can be updated every session.

2) PHYSICAL UNCLONABLE FUNCTIONS (PUF)

PUFs are hardware-level lightweight security mechanisms. PUF is one of the most commonly used primitives for IoT. PUF operates as a challenge-response mechanism, where the physical properties of a device serve as the *challenge*, and the resulting digital output is the *response*. This process generates a unique identifier for each device. Since the physical characteristics that form the basis of the PUF cannot be cloned or replicated, it ensures a high level of security.

This section summarizes several PUF-based protocols including their communication cost and authentication factors in Table 4. Table 5 compares their computation cost.

Mahalath et al. [11] propose a challenge-response protocol with a PUF mechanism and generate random keys as seeds. The study covers IoT devices that are connected to WiFi. Thus, security is provided between IoT devices and routers.

In another study, Braeken [59] propose a PUF-based key exchange protocol enhanced by the ECC method.

TABLE 2. Summary of ECC-based authentication protocols for IoT devices.

Study	Description	Main Focus	Factor(s)	Communication [Byte]
Liao and Hsiao, 2014 [20]	ECC-based RFID authentication protocol	Low time & computation overhead	RFID	82
Kalra and Sood, 2015 [6]	ECC-based Public Key Cryptographic mutual authentication	Computations cover XOR operations and hash functions	Password	160
Rostampour <i>et al.</i> , 2020 [56]	ECC-based authentication protocol for edge devices	Composes an ECC function that focuses solely on core elliptic curve operations, excluding other cryptographic primitives	–	N/A
Li <i>et al.</i> , 2021 [45]	OTP-based authentication and ECC-based key agreement protocol	Low computation cost	OTP	N/A
Iqbal <i>et al.</i> , 2022 [57]	ECC-based authentication and key exchange protocol between IoT devices and cloud server	Low computation power and energy consumption	–	176
Singh <i>et al.</i> , 2023 [53]	Anonymous ECC-based mutual authentication protocol between the users, the nodes, and the gateway	Low overhead and only XOR operations and hash functions are used	Password, Biometric	288
He <i>et al.</i> , 2023 [7]	ECC-based authentication and key exchange protocol with anonymity	Supported with timestamps and symmetric key encryption, short computation time	Password	82
Patel and Doshi, 2023 [8]	ECC-based authentication and level-dependent authorization protocol	Access permissions are managed with levels and short computation time	Password	320
Jayashree and Santhosh Kumar, 2023 [58]	Lightweight VANET authentication with signatures	Communication costs and time are decreased	–	231
Abbasi <i>et al.</i> , 2024 [54]	Cloud-enabled ECC-based healthcare system authentication protocol	Secure, fast, and lower computation and communication cost compared to similar studies	Password, Biometric	166
Saqib and Moon, 2024 [10]	ECC-based multi-factor protocol for MQTT-based IoT	Dynamic session key, low-cost signature	Password, Biometric	236

TABLE 3. Computation costs of ECC-based authentication protocols.

Study	User	Device/Sensor	Gateway	Server	Total
[6]	3SM	–	–	4SM	7SM
[56]	–	7SM+1PA	–	6SM+1PA	13SM+2PA
[45]	–	2SM+3H	–	2SM+3H	4SM+6H
[57]	–	4SM+1PA+2H+1ED+1INV	–	4SM+1PA+2H+1ED+1INV	8SM+2PA+4H+2ED+2INV
[53]	2PM+1H	3PM+2H+2ED	1PM+3H+2ED	–	6PM+6H+4ED
[7]	5H+2ED+1SM	–	–	6H+3ED+1SM	11H+5ED+2SM
[8]	6H+2ED	6H+3ED	13H+6ED	–	25H+11ED
[58]	–	3SM+1PA	–	1SM+1PA	4SM+2PA
[54]	1ECC+6H+1ED	–	–	2ECC+4H+2ED	3ECC+10H+3ED
[10]	4H+2SM+1F	1H+3SM	2H+5SM	–	7H+10SM+1F

H: hash/HMAC functions, ED: symmetric encryption or decryption, ECC: elliptic curve cryptography, SM: scalar multiplication, PA: point addition, PM: point multiplication, INV: multiplicative inverse, F: fuzzy extractor

*Authentication cost includes message signing and message verification costs in VANET studies.

Mahmod and Guin [60] propose a PUF-based authentication using static RAM (SRAM) information. The authors also

propose a repeated ID matching method including Hamming Distance method against repeated authentications.

Nimmy et al. [61] propose a distinct approach for challenge-response of PUF. The authors do not store challenge-response pairs in the verifier database. The geometric secret-sharing method is used to prevent the explicit storage of challenge-response pairs.

Yildiz et al. [62] propose a distinct study in terms of its scope. The study is a PUF-based authentication and key exchange protocol for groups of IoT devices. The protocol uses PUF, factorial tree, and Chinese Remainder Theorem (CRT) as underlying methods as well as XOR, hash functions, and symmetric encryption as cryptographic primitives.

Ebrahimabadi et al. [63] present a distinct approach to PUF-based authentication protocol. The authors propose a challenge-response mechanism named challenge splitting (CSP), wherein the server partitions the challenge into segments distributed between the device and auxiliary helper devices. The method is improved using scrambling and padding methods and machine learning techniques.

Yoon et al. [64] propose a novel study that generates PUF using static RAM information and hashing of amplitudes of channel state information. The authors try to prevent the exposure of challenge-response pairs generated with PUF. A hashing model and the encryption key are generated using machine learning algorithms. The CSI dataset is used for training.

Yu and Park [55] propose a PUF-based authentication and key exchange protocol that uses smart card, password, and biometric factors. The authors try to provide a cost-effective protocol between smart vehicles, charging stations, and service providers.

Saleem et al. [23] propose a PUF and ECC-based authentication for VANETs. The authors provide an efficient protocol for multiple vehicles.

Tanveer and Aldossari [66] propose an authentication protocol for 6G edge networks. The protocol includes ECC, PUF, and ASCON encryption cryptographic primitives. ASCON is an algorithm that provides encryption and authentication. The cost of the protocol is low according to similar studies.

B. AUTHENTICATION FACTOR

In this section, authentication factors are classified into four categories. The categories are formed based on the most frequently encountered approaches of the recent studies. The studies are evaluated based on authentication factors, communication costs, and computation costs.

1) ONE TIME PASSWORD AUTHENTICATION

The one-time password (OTP) is a dynamically changed password, which depends on pre-shared information between pairs, for every session, login attempt, or transaction. OTP effectively prevents many attacks that arise from reusable passwords.

Zhao et al. [47] propose an OTP-based authentication using the negative database method (NDB). NDB is the complement of the database, contributing to improved

security. NDB takes up a smaller space than an ordinary database.

Erdem et al. [48] provide a cloud-based OTP authentication between the user, an OTP provider, and a service provider. The authors offer unlinkable profiles to manage the OTP provider with different roles.

Hammi et al. [46] propose a lightweight OTP authentication that is based on ECC and isogeny methods. The authors provide a new approach using isogeny and require less power consumption to generate an OTP.

Su et al.'s study [49] includes an OTP-based authentication protocol with password and biometric authentication. The authors use lightweight cryptographic primitives and symmetric encryption. The computation time of the protocol is shorter than in similar studies.

Hayouni et al.'s [51] study proposes an OTP authentication mechanism based on challenge-response to enhance security. The authors focus on the security of WSNs. In their network topology, a main entity (like a gateway) is used for management with unlimited resources.

Lone and Mir [50] propose a different approach to OTP-based authentication. The authors use usernames, device numbers, and fingerprint information as the seed to generate the OTP. The success of the registration and authentication phases are measured.

Goel et al. [52] propose a distinct type of study on OTP authentication. The authors provide symmetric encryption between the IoT device and the server while they use a time-based OTP.

Table 6 summarizes several OTP-based authentication protocols. Table 7 compares the studies using their computation cost.

2) SMART CARD AUTHENTICATION

Smart cards contain embedded integrated circuits and are used to control access. It is one of the most preferred possession factors due to its user-friendly nature. This section presents numerous studies that explore smart card authentication.

Das et al. [35] propose a two-factor authentication protocol to improve the security of electronic patient record systems. The authors' study is computationally cost-effective, as they use hash functions and bit-wise XOR operations throughout the entire protocol.

Amin et al. [37] propose a password-based smart card authentication for multi-server cloud. The proposed method utilizes hash functions, XOR operations, and timestamps to enhance security.

Kou et al.'s [38] authentication study includes all three factors as the smart card, the password, and the biometric data. The protocol takes the password and biometrics to compute a unique value by utilizing fuzzy extractor functions.

Another smart card authentication study that proposes two-factor authentication is by Rana et al. [39]. The study uses lightweight cryptographic primitives and timestamps through the entire protocol.

TABLE 4. Summary of PUF-based authentication protocols for IoT devices.

Study	Description	Main Focus	Combined Method/Factor(s)	Communication [Byte]
Mahalat <i>et al.</i> , 2018 [11]	PUF-based challenge-response authentication protocol	Only XOR operations, hash functions, and PUF calculations with nonces	Password	N/A
Braeken, 2018 [59]	PUF-based key agreement protocol with ECC	Fast protocol with low overhead	–	192
Mahmod and Guin, 2020 [60]	SRAM-PUF authentication protocol	Repeated ID matching scheme to enhance PUF security	–	N/A
Nimmy <i>et al.</i> , 2021 [61]	PUF-based authentication and key exchange protocol	PUF-based challenge-responses are stored in hashed form	–	252
Yildiz <i>et al.</i> , 2021 [62]	Group authentication and key distribution protocol for (de)centralized architectures	Reduced number of communication messages	–	N/A
Ebrahimabadi <i>et al.</i> , 2022 [63]	PUF-based authentication protocol against modeling-attack	Strong algorithm supported with challenge splitting, padding, scrambling, and machine learning	–	N/A
Yoon <i>et al.</i> , 2023 [64]	Joint heterogeneous PUF-based authentication protocol	Challenge-responses are generated using physical information and machine learning algorithms are used to generate robust keys and hashing model	–	27
Yu and Park, 2024 [55]	PUF-based authentication and key exchange protocol for the vehicle to grid networks	All computations are low-cost as only XOR operations and hash functions are used	Smart card, password, biometric	245
Saleem <i>et al.</i> , 2024 [23]	VANET authentication with PUF method	High efficiency for multi-vehicle communications	–	946
Tanveer and Aldossari, 2025 [66]	Mobile edge network authentication with PUF and ECC	Low cost and robust against several attacks	–	204

TABLE 5. Computation costs of PUF-based authentication protocols.

Study	User/Device	Server/Router/Verifier	Total
[11]	1H	1R	1H+1R
[59]	5H+3SM+1PA	10H+1SM+2PA	15H+4SM+3PA
[61]	5PUF+17H	8H+2R	5PUF+25H+2R
[62]	2H+2PUF+3ED	2H+4ED+nR	4H+2PUF+7ED+nR
[55]	2R+9H	15H+5R+4ED	7R+24H+4ED
[23] ¹	(3SM+3H+3PA)+(3SM+2H+3PA)	2PUF	6SM+5H+6PA+2PUF
[66]	5H+1PUF+5AS+1F+2SM	1PUF+4AS+1F+1SM	10H+2PUF+3SM+2F+9AS

H: hash/HMAC functions, ED: symmetric encryption or decryption, PUF: physically unclonable functions, SM: scalar multiplication, F: fuzzy extractor, AS: ASCON algorithm, PA: point addition, R: random number generation

¹ The authentication occurs between the vehicles, the device computation cost covers two vehicles' computation costs in the User/Device column.

Soni and Singh [40] propose a lightweight authentication and key agreement protocol based on smart cards and passwords for the Wireless Body Area networks. The protocol uses only lightweight cryptographic primitives.

Kumar *et al.* [41] propose a smart card authentication using the biometric fuzzy extractor method. The study also defines an extra revocation step.

Cherbal and Benchetioui [34] propose an ECC-based smart card authentication together with passwords.

Another approach for smart card authentication is by Khan *et al.* [76]. They propose a two-factor authentication including smart card and password authentication factors for health systems. The authors use hyperelliptic curve cryptography.

Alizadeh *et al.* [42] propose a smart card authentication with identity aliases to provide anonymity. The authors propose a ticket-based method for strong authentication. The ticket is generated by the authentication server and includes

TABLE 6. Summary of OTP-based authentication protocols for IoT devices.

Study	Description	Main Focus	Factor(s)	Combined Method/Factor(s)	Communication [Byte]
Zhao and Luo, 2017 [47]	OTP-based authentication protocol	NDB usage as an extra layer to improve the security	Two	Password	N/A
Erdem and Sandikkaya, 2018 [48]	Cloud-based OTP authentication protocol	Each user has own profile that are independent of others	Two	Password	N/A
Hammi <i>et al.</i> , 2020 [46]	OTP-based lightweight authentication for IoT and smart city applications	Less power consumption with a different type of OTP generation	One	ECC, isogeny	N/A
Su <i>et al.</i> , 2021 [49]	OTP-based multi-factor authentication protocol	Low computation overhead and strong authentication	Three	Biometric, password	420
Lone and Mir, 2022 [50]	OTP-based multi-factor authentication protocol	Robust authentication, low computation time	Two	Biometric seed	N/A
Hayouni, 2022 [51]	OTP-based authentication protocol for WSNs	Low overhead and distinct network topology	One	–	N/A
Goel <i>et al.</i> , 2022 [52]	Time-based OTP authentication with symmetric key encryption	Low overhead for the encryption algorithm	One	Symmetric key encryption	N/A

TABLE 7. Computation costs of OTP-based authentication protocols.

Study	User	Device/Sensor	Gateway	Server	Total
[47]	1R+1H+1NDB	–	–	1R+1H+1NDB	2R+2H+2NDB
[48]	1R (OTP)	–	–	–	1R (OTP)
[49]	7H+2ED+2SM	4H+2ED	9H+4ED+1SM	–	20H+8ED+3SM
[46]	–	1R+2SM+2IS+1H	–	1R+2SM+2IS+1H	2R+4SM+4IS+2H
[51]	–	3H	1R+3H	–	1R+6H

NDB: negative database, H: hash function, R: random number generation, SM: scalar multiplication, ED: symmetric encryption or decryption, IS: isogeny

information about the expiry time and IoT device identity. It should be valid to proceed the authentication process.

Table 8 summarizes several smart card authentication protocols. Table 9 compares the studies using their computation cost including functions.

3) PASSWORD-BASED AUTHENTICATION

Password authentication is the most familiar authentication method for users. Therefore, many authors try to strengthen password authentication with various cryptographic approaches.

Renuka *et al.* [12] propose an authentication protocol for machine-to-machine networks. The protocol is a password-based protocol including hash functions, XOR operations, and symmetric encryption.

Son *et al.* [13] perform cryptanalysis on a target study [77] and address its security deficiencies. The authors propose a new, secure, and low-cost authentication protocol that utilizes only hash functions and XOR operations.

Vinoth *et al.* [14] propose a password-based, biometric, and smart card multi-factor authentication protocol. The protocol includes hash functions, XOR operations, and symmetric encryption.

Masud *et al.* [15] present a lightweight user authentication protocol for real-time healthcare systems. The study uses only XOR operations and hash functions. The authors focus on cost-effectiveness at the terminal device while they try to present a protocol that is robust against common attacks.

Chen *et al.* [16] propose a distinct approach. The authors present a lightweight authentication protocol based on PUFs method and the firmware integrity of IoT devices as an authentication factor. The firmware is hashed to maintain device integrity.

Cetintav and Sandikkaya [17] propose a lightweight authentication, key exchange, and data transfer protocol including only hash functions and XOR operations. The authors present a distinct infrastructure for communication between hierarchically organized servers and IoT terminals. The ephemeral encryption keys are updated each transaction in between each participant. The study is secure against common attacks. Its computation and communication costs are low.

A two-factor authentication protocol is proposed by Han *et al.* [18]. The authors present a protocol based on password and biometric authentication factors.

TABLE 8. Summary of smart card authentication protocols for IoT devices.

Study	Description	Main Focus	Factor(s)	Combined Method/Factor(s)	Communication [Byte]
Das, 2015 [35]	A password and smart card authentication protocol	Only hash functions and XOR operations are used	Two	Password	–
Farash <i>et al.</i> , 2016 [36]	Smart card-based authentication and key exchange protocol	Low communication cost for each participant	Two	Password	434
Amin <i>et al.</i> , 2018 [37]	Authentication protocol for distributed cloud servers	Only hash functions and XOR operations are used	Two	Password	N/A
Kou <i>et al.</i> , 2019 [38]	Lightweight three-factor authentication	Multi-factor and lightweight	Three	Password, biometric	N/A
Rana <i>et al.</i> , 2021 [39]	Lightweight smart card authentication	Only hash functions and XOR operations are used	Two	Password	292
Soni and Singh, 2022 [40]	Lightweight authentication and key exchange protocol for WBAN	Only hash functions and XOR operations are used	Two	Password	180
Kumar <i>et al.</i> , 2023 [41]	Authenticated key exchange protocol	Robustness against several attacks	Three	Password, biometric	476
Cherbal and Benchetioui, 2023 [34]	ECC based smart card authentication	Computationally low cost	Two	Password, ECC	N/A
Alizadeh <i>et al.</i> , 2023 [42]	Smart card based two-factor authentication	Ticket-based authentication and anonym aliases	Two	Password	N/A

TABLE 9. Computation costs of smart card authentication protocols.

Study	User	Gateway	Device	Server	Total
[35]	8H	–	–	6H	14H
[36]	11H+1R	14H	7H+1R	–	32H+2R
[37]	9H+1R	–	–	14H+2R	23H+3R
[38]	7H + 1ED + 1F	2ED + 12H	–	–	19H + 2ED + 1F
[39]	20H	–	–	5H + 2ED	25H + 2ED
[40]	8H + 1R	–	–	5H + 1R	13H + 2R
[41]	3F + 12H + 3ED	5ED + 6H	2ED + 2H	–	3F + 20H + 10ED
[42]	–	6ED + 2H	8H + 1R + 3ED	4H + 1R + 2ED	14H + 11ED + 2R
[34]	5H + 4ECC	–	–	2H + 3ECC	7H + 7ECC

H: hash/HMAC functions, F: fuzzy extractor, ED: symmetric encryption or decryption, ECC: elliptic curve cryptography, R: random number generation

Another multi-factor authentication protocol is proposed by Saleem *et al.* [9]. The authors present a password and biometry-based authentication protocol using ECC and fuzzy extractors.

Table 10 summarizes several password-based authentication protocols. Table 11 compares the studies' computation costs.

4) RFID AUTHENTICATION

RFID is a frequently used technology, mostly as the possession factor, in authentication protocols. RFID technology requires an RFID tag/card and an RFID reader. Several IoT authentication studies utilize RFID as it is cheap and does not require user attention.

Kai *et al.* [24] propose a cloud-based RFID authentication for healthcare systems. The authors use quadratic residuals and pseudo-random number generators to improve security. The index data table stores the identities of tags and readers. Indexes and relevant secret keys are anonymous.

Gao *et al.* [29] propose an RFID-based Rabin public key encryption authentication protocol. The authors present an effective authentication protocol by enhancing the Rabin public key algorithm. As a distinct approach, the cost of the protocol is evaluated based on logic gates that are used for operations (i.e. multiplication operation).

Gope *et al.* [25] propose an RFID authentication for distributed IoT devices in smart city systems. The authors add secure localization to the RFID authentication method to provide untraceability.

Fan *et al.* [26] propose an RFID authentication protocol for medical devices. The authors use an index data in the server for effective data retrieval from the server. Cross operation¹ generates index value and the rotation operation generates index content. These operations provides verification to generate seeds.

¹Cross operation includes several sub-operations such as cascade and concatenation to generate an index value.

TABLE 10. Summary of password-based authentication protocols for IoT devices.

Study	Description	Main Focus	Factor(s)	Combined Method/Factor(s)	Communication [Byte]
Renuka <i>et al.</i> , 2019 [12]	Low cost authentication protocol for M2M networks	The encryption cost is almost equal to hash cost	One	–	N/A
Son <i>et al.</i> , 2021 [13]	Lightweight anonymous authentication	Only hash functions and XOR operations are used	One	–	200
Vinoth <i>et al.</i> , 2021 [14]	Multi-factor authentication including password, biometric, and smart card factor	Scalable	Three	Biometric, smart card	380
Masud <i>et al.</i> , 2022 [15]	Password-based authentication and key agreement protocol	Low computation cost using only hash functions and XOR operations	One	–	256
Chen <i>et al.</i> , 2023 [16]	Firmware integrity and PUF based authentication protocol	A new and distinct lightweight approach with firmware integrity	Two	Firmware integrity	132
Cetintav and Sandikkaya, 2023 [17]	Lightweight authentication and key exchange	Low cost	One	–	352
Han <i>et al.</i> , 2024 [18]	Two factor authentication with password and biometry	Two generated keys form master key to provide security	Two	Biometric	1220
Saleem <i>et al.</i> , 2024 [9]	Anonymous authentication and key exchange protocol for VANETs		Two	Biometric, ECC	164

TABLE 11. Computation costs of password authentication protocols.

Study	User	Gateway/RSU	Device	Server	Total
[15]	3H	3H	2H	–	8H
[12]	5ED	11ED	11ED	–	27ED
[13]	9H	–	–	12H	21H
[14]	9H+1F+1ED	6H+3ED+2MM	4H+2ED+1Mod	–	19H+6ED+2MM+1Mod+1F
[16]	–	–	5H+1PUF	4H+1F	9H+1PUF+1F
[17]	7H	–	10H	7H	24H
[9]	–	4H+1F	3H+2ED	–	9H+10SM

H: hash/HMAC functions, F: fuzzy extractor, ED: symmetric encryption or decryption, MM: modular multiplication, Mod: modular arithmetic, SM: scalar multiplication

Aghili *et al.* [27] propose an enhanced authentication protocol of Fan *et al.*'s [26] protocol using a modular rotation method. The authors focus on the healthcare IoT devices that use RFID technology on the medical sensors.

Fan *et al.* [28] propose an RFID-based authentication using pseudo-random number generators and shift operations.

Zhu *et al.* [30] propose an RFID-based lightweight authentication for healthcare systems. The authors use lightweight cryptographic primitives such as pseudo-random number generators and hash functions. The tag has a low computational cost.

Ji *et al.* [31] propose an RFID authentication protocol for transportation systems. The authors use two data tables to store the information about tags and readers. The proposed protocol has low computational and communication costs.

Another approach is proposed by Hosseinzadeh *et al.* [32]. The authors provide an RFID-based authentication using

the Rabin public-key protocol against desynchronization and traceability attacks.

Gabsi *et al.* [22] propose an ECC-based RFID authentication protocol. The authors use scalar multiplication, point addition, and random number generation methods to provide lightweight security.

Izza *et al.* [21] propose an RFID authentication using the ECC public key cryptography method. The authors provide an ECC-based digital signature with message recovery to detect security issues and transmit large data.

Vijaykumar *et al.* [33] propose a distinct approach to RFID authentication by designing a multifunction logic circuit. The authors provide security with logical outputs generated for each random selection of logical inputs.

Saleem *et al.* [23] propose an RFID authentication for VANETs. The authors use the PUFs for challenge-response

TABLE 12. Summary of RFID-based authentication protocols for IoT devices.

Study	Description	Main Focus	Factor(s)	Combined Method/Factor(s)	Communication [Byte]
Fan <i>et al.</i> , 2016 [24]	Cloud-based RFID authentication protocol	Tags have low computational cost	One	–	N/A
Gope <i>et al.</i> , 2018 [25]	Anonymous RFID authentication protocol	Mutual authentication and strong anonymity	One	–	N/A
Fan <i>et al.</i> , 2018 [26]	RFID-based authentication for medical privacy	Strong security protocol with low computational cost for tags	One	–	N/A
Aghili <i>et al.</i> , 2019 [27]	RFID-based authentication protocol for the health systems	Enhancing the security using modular rotation function	One	–	N/A
Fan <i>et al.</i> , 2019 [28]	Cloud-based RFID authentication for healthcare systems	Lightweight cryptographic primitives and low computation complexity	One	–	N/A
Gao <i>et al.</i> , 2019 [29]	Rabin public key encryption based RFID authentication protocol	Implementation requires less number of gates than similar studies	One	Rabin public key encryption	N/A
Zhu <i>et al.</i> , 2020 [30]	RFID authentication protocol for healthcare systems	Low computational cost on smart devices	One	–	168
Ji <i>et al.</i> [31] (2020)	Anonymous RFID-based authentication protocol for VANET	Lightweight cryptographic primitives, low communication and storage cost	One	–	N/A
Hosseinizadeh <i>et al.</i> , 2020 [32]	RFID-based authentication with Rabin public key	Low communication cost and strong public key method	One	Rabin public key encryption	160
Gabsi <i>et al.</i> , 2021 [22]	ECC-based RFID authentication protocol	Low cost for tags	One	ECC	160
Izza <i>et al.</i> , 2021 [21]	RFID-based authentication and data transmission protocol for WBAN	ECC and ECC-based signature usage	One	ECC	31
Vijaykumar <i>et al.</i> , 2024 [33]	Hardware-based RFID authentication protocol	Low energy consumption	Two	Password	25
Saleem <i>et al.</i> , 2024 [23]	RFID-based authentication protocol with PUF method	Low overhead on vehicles	One	PUF	262

TABLE 13. Computation costs of RFID-based authentication protocols.

Study	Tag/Device	Reader/RSU	Server	Total
[24]	R+2H	R+H	–	2R+3H
[25]	5H	2H	7H	14H
[26]	3Cro+1P	7Cro+1P+1Rot	6Cro+1P+1Rot	16Cro+2Rot+3P
[27]	1R+7M	1R+17M	1R+5M	3R+29M
[28]	3P+2Rot	5P+5Rot+3M+1S	3Rot+5P	13P+10Rot+3M+1S
[29]	4H+1R	2H+2R	–	6H+3R
[30]	5P	5P	9P	19P
[31]	1R+5Rot	1R+5H	1R+5Rot+5H	3R+10Rot+10H
[32]	1MSO+1R+2H	1SR+1R+2H	–	1MSO+2R+4H+1SR
[22]	1R+4SM+2PA	1R+4SM+2PA	–	2R+8SM+4PA
[21]	4SM+6H	4SM+10H	3SM+4H	11SM+20H
[33]	1R+2MFC	1R+2MFC	1R+2MFC	3R+6MFC
[23]	3H+1PUF+2HM	–	3H+2ED+2HM	5H+1PUF+4HM

Cro: cross operation, P: pseudo-random number generator, R: random number generator, Rot: rotation operation, M: modular rotation operation, MFC: multifunction circuit operation, SM: scalar multiplication, PA: point addition, MSO: modulo squaring operation, H: hash/HMAC functions, SR: squaring root, HM: henon map function, ED: symmetric encryption or decryption

pairs and the Henon map shuffling function as a distinguishing approach. The cost and overhead are reduced using lightweight cryptographic primitives.

Table 12 summarizes several RFID-based authentication studies. Table 13 compares the selected RFID-based protocols' computation costs.

TABLE 14. Domain-specific summary of authentication protocols for IoT.

Study	Description	Authentication Factor(s)	Domain
[44]	A robust authentication protocol for medical WSNs	Password & smart card	Medical WSN
[67]	A lightweight remote authentication method for Wireless Body Area Networks	–	WBAN & Healthcare
[19]	A lightweight user authentication scheme for patient monitoring	Password	Healthcare
[68]	ECC-based password authentication	–	Distributed IoT
[69]	A lightweight authentication for anonymity	–	WBAN
[70]	A hash and certificate-based lightweight authentication and key agreement protocol for drones	–	UAV
[71]	Chebyshev chaotic mapping and ECC-based authentication protocol	–	UAV
[43]	A lightweight authentication and key agreement protocol	Smart card & password	Healthcare
[72]	A lightweight signature-based authentication	–	Smart home
[73]	An authentication and key agreement protocol with certificateless public key cryptography	–	Smart grid
[9]	A lightweight password and biometric authentication protocol	Password & biometric	VANETs
[75]	A symmetric encryption based authentication protocol	–	Internet of medical things
[58]	Lightweight VANET authentication with signatures	–	VANETs
[54]	Cloud-enabled ECC-based healthcare system authentication protocol	Passwords & biometric	Healthcare
[51]	OTP-based authentication protocol	OTP	WSN
[40]	Lightweight smart card authentication and key exchange protocol	Smart card & password	WBAN
[12]	Low cost password-based authentication protocol	Password	M2M networks
[15]	Password-based authentication and key agreement protocol	Password	Healthcare
[23]	RFID-based lightweight authentication protocol	RFID	VANETs
[24]	Cloud-based RFID authentication protocol	–	Healthcare
[25]	RFID-based lightweight authentication protocol	RFID	Smart city
[27]	RFID-based lightweight authentication protocol	RFID	Healthcare
[30]	RFID-based lightweight authentication protocol	RFID	Healthcare
[31]	Anonymous RFID-based authentication protocol	–	VANETs
[21]	RFID-based authentication and data transmission protocol	–	WBAN

C. AUTHENTICATION STUDIES BY APPLICATION DOMAINS

Table 14 summarizes studies that are used in different domains. The studies in the table indicate a specific domain, unlike the others. Healthcare and vehicular networks are significant areas of deployment but still require optimized protocols for resource constraints.

D. STRENGTH OF PRESENTED STUDIES AGAINST COMMON ATTACKS

Reviewed studies are evaluated based on their resistance against common attacks. Table 15 presents the attack resistance claims of the relevant studies. A checkmark indicates that the corresponding protocol successfully defends against the specified attack.

TABLE 15. The resistance of the reviewed studies against various attacks is analyzed.

Study	E	DoS	R	IN	D	BF	PS	IM	TA	MitM	FS	OG	F
[31]			✓								✓		
[39]		✓	✓	✓				✓		✓		✓	
[8]			✓					✓		✓	✓		
[74]				✓								✓	✓
[78]		✓	✓					✓					
[79]			✓										✓
[80]			✓	✓				✓		✓		✓	
[6]	✓		✓		✓	✓				✓	✓		
[81]		✓									✓		
[60]		✓	✓										
[62]	✓		✓					✓		✓	✓		
[82]			✓		✓					✓			
[20]		✓	✓								✓		
[83]			✓					✓					
[24]	✓	✓	✓								✓		
[84]		✓	✓				✓	✓		✓	✓	✓	
[85]			✓					✓					
[86]		✓	✓					✓					
[44]				✓				✓				✓	
[19]		✓	✓	✓				✓		✓	✓	✓	
[53]		✓	✓					✓		✓		✓	
[58]			✓							✓			
[54]		✓	✓					✓		✓	✓	✓	✓
[61]		✓	✓					✓		✓	✓		
[64]			✓			✓				✓			
[55]			✓					✓		✓	✓	✓	
[47]			✓							✓		✓	
[48]		✓	✓	✓						✓			✓
[49]			✓					✓					
[52]			✓			✓							
[35]			✓	✓				✓		✓		✓	
[36]		✓	✓	✓				✓		✓		✓	
[37]			✓	✓				✓				✓	

TABLE 15. The resistance of the reviewed studies against various attacks is analyzed, continued.

Study	E	DoS	R	IN	D	BF	PS	IM	TA	MitM	FS	OG	F
[38]		✓	✓	✓				✓		✓		✓	
[40]			✓					✓		✓			
[41]		✓		✓				✓			✓		
[34]		✓	✓				✓			✓		✓	
[42]		✓	✓					✓				✓	✓
[25]		✓	✓								✓		✓
[26]		✓	✓								✓		
[27]			✓					✓			✓		
[30]								✓			✓		
[32]			✓					✓					
[22]		✓	✓					✓			✓		
[21]			✓	✓				✓			✓		
[68]			✓							✓			
[15]		✓	✓	✓				✓		✓		✓	
[65]			✓					✓			✓		
[13]				✓				✓		✓	✓	✓	
[16]			✓					✓		✓	✓		
[17]		✓	✓					✓		✓	✓		
[18]			✓		✓			✓				✓	
[23]								✓			✓		
[9]					✓	✓		✓			✓	✓	
[57]			✓					✓		✓			
[63]	✓		✓					✓					
[51]		✓	✓			✓							
[66]		✓	✓	✓				✓		✓			
[56]			✓							✓			
[10]			✓	✓				✓		✓	✓	✓	

Eavesdropping (E), Denial of Service (DoS), Replay Attack (R), Insider Attack (IN), Dictionary attack (D), Brute Force attack (BF), Parallel Session Attack (PS), Impersonation (IM), Traffic Analysis (TA), Man-in-the-middle Attack (MitM), Forward Secrecy (FS), Offline Guessing (OG), Forgery (F)

The short descriptions of the presented attacks and features are given below.

Eavesdropping (E): An attacker observes the communication between participants (i.e. device and server) and gets sensitive information such as ID and passwords.

Denial of Service (DoS): Attackers attempt to disrupt services or devices by sending numerous requests. The excessive traffic could render services unavailable, leading to financial damage.

Replay attack (R): Attackers intercept transmitted messages and replay them to the receiver at a later time

TABLE 16. Overview of related works: similarities and differences between this study and the literature are shown.

Study	Similarities	Differences	Classification Notes
[87]	Classification based on resistances, cryptographic operations, and target domains, potential attacks, countermeasures, security weaknesses and strengths, cost analysis with computation, and communication overheads are examined	Security verifications are shown. Distinct attack types are included.	–
[88]	Akin to this paper, the authors classify cost, security threats, and claimed benefits	The authors classify the authentication schemes into different classes	–
[89]	Authentication methods, strengths and weaknesses, cost analysis	Detailed attack classification and the reviews are based on the attacks. Formal security comparison, evaluation techniques	–
[90]	–	Attack-based classification of the authentication protocols	They classify authentication schemes as identity-based, cryptography-based, token-based, hardware-based, procedure-based, context-based
[91]	–	Authentication type classification. They classify a set of lightweight authentication protocols into four categories	Public key-based, token-based, procedural-based, hardware-based
[92]	Challenges and weaknesses are mentioned together with performance analysis	Research method is based on different authentication type classification	Cloud-based, lightweight, decentralized blockchain-based, biometrics-based remote authentication
[93]	–	Based on a distinct taxonomy	Context awareness, topology, and communication type are among the focused properties
[96]	–	For the paper selection, the main focus is the integrity of IoT applications	Focused layer, cryptographic method, strength and weaknesses
[94]	Application domains, authentication factors	IoT layers, architecture, token usage	Comprehensive analyse and classification
This paper	–	–	Lightweight authentication classification based on security requirements, authentication factor, usage areas, and cryptographic primitives. Attack resistance, computation, communication, and hardware cost correlation.

or in a subsequent session, posing as a legitimate participant.

Insider attack (IN): An insider attack differs in terms of the attacker. It involves a participant who has access to system information and intentionally or unintentionally becomes the attacker.

Dictionary attack (D): Attackers attempt to reveal passwords or cryptographic keys by trying all possible options from a previously compiled dictionary list.

Brute force attack (BF): Attackers automatically attempt to find the passwords or keys by trying every possible character combination.

Parallel session attack (PS): While a legitimate user is still using the services, an attacker impersonates a legitimate

user for unauthorized access to the services by creating a session in parallel. For this attack, attackers should get user credentials or other necessary information to create a session.

Impersonation (IM): An attacker roles as a legitimate user/device to access the system/services.

Man-in-the-middle attack (MitM): Attackers are located in the middle of the communication channel that is between legitimate participants. The attackers listen (a type of eavesdropping) to the traffic, transferred data, and user/device information; and steal and manipulate them.

Forward secrecy attack (FS): Messages eavesdropped during protocol execution could be cryptanalyzed

TABLE 17. A comprehensive summary of IoT authentication protocols is provided, focusing on their resistance to various types of attacks, as well as their computational, communication, and hardware costs. Computation and communication costs are total.

Study	Attack Resistance	Computation Cost	Communication Cost [Bytes]	Hardware Cost
[6]	E,RA,D,BF,MitM,FS	7SM	160	-
[57]	RA,IM,,MitM	8SM+2PA+4H+2ED+2INV	176	-
[53]	DoS,RA,IM,MitM,OG	6PM+6H+4ED	280	1(BIO)
[7]	N/A	11H+5ED+2SM	82	-
[8]	RA,IM,MIMT,FS	25H+11ED	320	-
[58]	RA,MitM	4SM+2PA	231	-
[54]	DoS,RA,IM,MitM,FS,OG,F	3ECC+10H+3ED	166	1(BIO)
[59]	N/A	15H+4SM+3PA	192	1(PUF)
[61]	DoS,RA,IM,MitM,FS	5PUF+25H+2R	252	1(PUF)
[55]	RA,IM,MitM,FS,OG	7R+24H+4ED	245	2(SC)+1(BIO)+1(PUF)
[65]	RA,IM,FS	6SM+5H+6A+2PUF	946	1(PUF)
[49]	RA,IM	20H+8ED+3SM	420	1(BIO)
[36]	DoS,RA,IN,IM,MitM,OG	32H+2R	434	2(SC)
[39]	DoS,RA,IN,IM,MitM,OG	25H+2ED	292	2(SC)
[40]	RA,IM,MitM	13H+2R	180	2(SC)
[41]	DoS,IN,IM,FS	2ED+2H	476	2(SC)+1(BIO)
[13]	IN,IM,MitM,FS,OG	21H	200	-
[14]	N/A	19H+6ED+2MM+1Mod+1F	380	1(BIO)+2(SC)
[16]	RA,IM,MitM,FS	9H+1PUF+1F	132	-
[9]	D,BF,IM,FS,OG	9H+10SM	164	1(BIO)
[30]	IM,FS	19P	168	2(RFID)
[32]	RA,IM	1MSO+2R+4H+1SR	160	2(RFID)
[22]	DoS,RA,IM,FS	2R+8SM+4PA	160	2(RFID)
[21]	RA,IN,IM,FS	11SM+20H	248	2(RFID)
[33]	N/A	3RA+6MFC	25	2(RFID)
[23]	IM,FS	5H+1PUF+4HM	262	2(RFID)+1(PUF)
[17]	DoS,RA,IM,MitM,FS	24H	352	-
[66]	DoS,RA,IN,IM,MitM	10H+2PUF+3SM+2F+9AS	204	1(PUF)

H: hash/HMAC functions, F: fuzzy extractor, ED: symmetric encryption or decryption, ECC: elliptic curve cryptography, SM: scalar multiplication, PA: point addition, PM: point multiplication, R: random number generation, INV: multiplicative inverse, MM: modular multiplication, P: pseudo-random number generator, Rot: rotation operation, M: modular rotation operation, MFC: multifunction circuit operation, MSO: modulo squaring operation, SR: squaring root, HM: henon map function

Eavesdropping (E), Denial of Service (DoS), Replay Attack (RA), Insider Attack (IN), Dictionary attack (D), Brute Force attack (BF), Parallel Session Attack (PS), Impersonation (IM), Traffic Analysis (TA), Man-in-the-middle Attack (MitM), Forward Secrecy (FS)*, Offline Guessing (OG), Forgery (F)

afterwards with the help of information acquired after the protocol's execution

Offline guessing (OG): Attackers attempt to find passwords instead of guessing passwords in real-time. Attackers are not constrained by time limits when attempting to guess passwords, thus they have an advantage, unlike in online guessing.

Forgery attack (F): An attacker sends fake data/e-mail/document creating a fake data/e-mail/document or altering data/e-mail/document for unauthorized access.

IoT devices are affected by the selected attack types, and most of the security studies aim to present protection against these attacks. This study reviews the literature studies based on the mentioned attacks. The selected studies may also focus on different types of attacks besides the mentioned attacks.

IV. RELATED WORKS

Surveys that exist in the literature classify IoT authentication studies from different perspectives. Table 16 shows the

comparison of the related works and this review. Most studies focus on a small group of papers and offer up-to-date reviews. Rao and Deebak [87] classify authentication and key management methods and compare their performances and security issues. Saqib and Moon [88] evaluate the authentication schemes in the IoT environment security based on attacks and classify authentication types. The authors also review the studies' formal verification techniques. Nandy et al. [89] review authentication protocols in terms of their robustness against each attack and utilized authentication methods. Szymoniak and Kesar [90] summarize the IoT authentication protocols based on cryptographic methods and security levels against common attacks. Lucia et al. [91] shortly summarize the IoT authentication protocols based on authentication factors. Kavianpour et al. [92] explain security issues and challenges of IoT authentication protocols. The authors compare the studies based on attack mitigation while summarizing their application domain and user interaction types (local/remote authentication). Trnka et al. [93] provide a taxonomy for the authentication and authorization protocols.

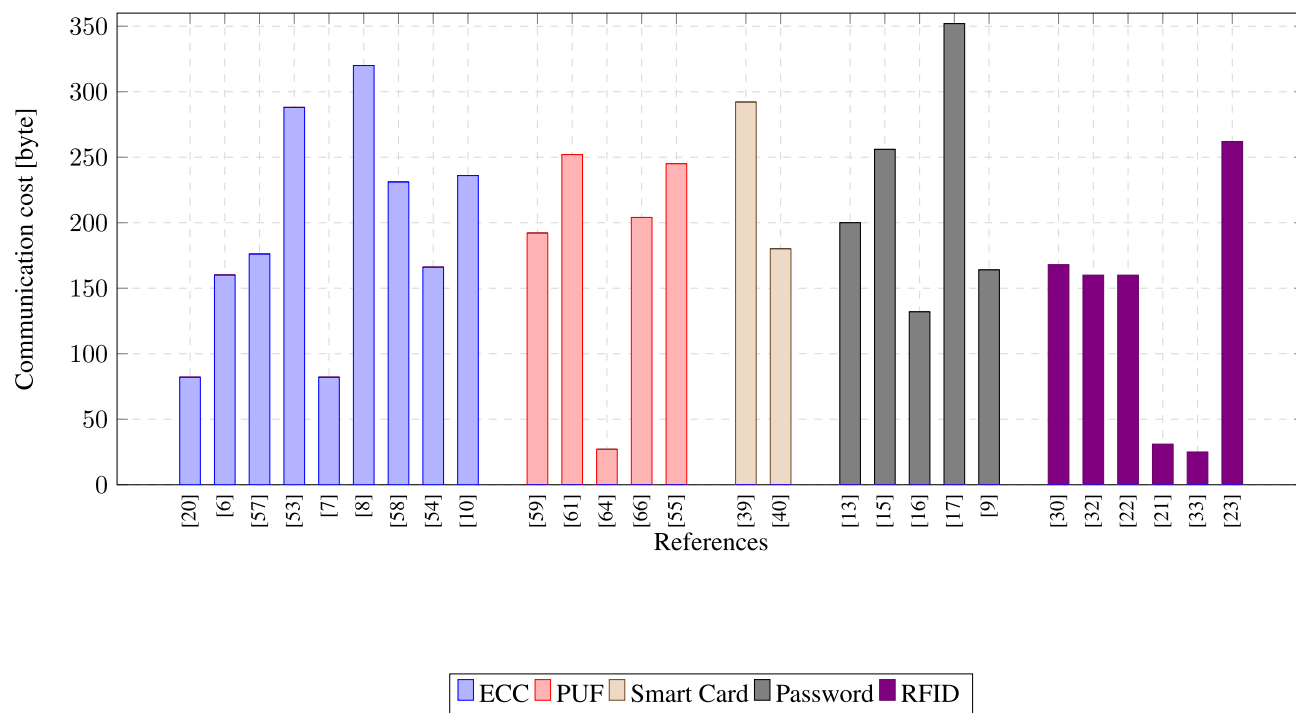


FIGURE 3. Comparison of the studies' communication costs.

The authors comprehensively summarize the studies with several properties such as communication methods, network topologies, and authentication factors. El-Hajj et al. [94] classified IoT protocols based on several categories such as architecture, factors, application domains, and token usages. Weaknesses and strengths are presented. Kumar et al. [95] examine the studies based on the application domains with advantages and disadvantages. The authors also presents the authentication factors with strengths against attacks. This study presents a comprehensive review for IoT authentication protocols attack resiliencies and cost of preferred methods. Some existing studies provide cost analyses but do not focus on the relationship between costs and security achievements. This review stands out from previous studies by addressing attack resistance, application domains, and providing a comprehensive cost analysis.

V. CONCLUSION

This paper reviews recent IoT authentication protocols by classifying them based on authentication features and costs. Table 17 summarizes studies with respect to their attack resistance, computational cost, communication cost, and additional hardware requirements. This allows the differences, advantages, and disadvantages of the studies to be easily identified. The *Attack Resistance* column indicates the claimed resistance against a set of attacks, while the *Hardware Cost* column specifies the required additional hardware to implement the protocol. Common off-the-shelf

devices, such as smartphones, tablets, and computers, are not considered additional hardware.

The systematic review categorizes IoT authentication studies based on two key factors: the underlying cryptographic approach and the utilized authentication factors. The most commonly encountered primitives are listed in the subcategories of ECC and PUFs, while the most frequently used authentication factors include smart cards, RFID tags, passwords, and OTPs.

Figure 3 shows the comparison of the reviewed studies' communication costs. The comparison of communication costs across different methods (ECC, PUF, Smart Card, Password, and RFID) reveals notable trends. According to the selected papers, ECC generally offers medium to high costs, balancing security with efficiency, while PUF stands out as a cost-effective solution with consistently low to medium costs. Smart card studies show moderate costs, indicating a balanced cost-performance ratio. In contrast, password-based methods exhibit higher costs. RFID studies display highly variable costs, ranging from low to very high, depending on the specific application context. The reviewed OTP authentication papers generally report high communication costs or lack information regarding communication costs. These findings highlight the importance of selecting a method that aligns with security requirements, available resources, and implementation needs, as performance and costs can vary significantly depending on the context. For instance, Kalra and Sood study [6] demonstrates one of the lowest

communication and computation costs, requires no additional hardware, and resists several types of attacks. Similarly, Cetintav and Sandikkaya study [17] is robust against multiple attacks, achieves one of the lowest computation costs, and also requires no additional hardware. In the literature, protocols that are low-cost, computation-efficient, and provide high security without requiring additional hardware are relatively rare, especially for resource-constrained IoT devices.

The application domain of IoT authentication varies based on its use case and purpose. In recent years, the remote control of IoT devices has become increasingly crucial. However, the need for additional hardware can pose challenges due to resource constraints. This study reveals that several approaches involve additional hardware and often require local user interaction.

Future work could explore artificial intelligence implementations designed to enhance IoT device security. This includes investigating machine learning models for real-time threat detection, optimizing resource allocation in lightweight authentication protocols, and dynamically adjusting cryptographic parameters based on device and network contexts. Furthermore, AI-driven self-healing authentication mechanisms that adapt to evolving threats and scalable solutions for diverse IoT environments, such as healthcare, smart cities, and industrial IoT, offer promising directions for addressing current challenges while ensuring resource efficiency and robust security.

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